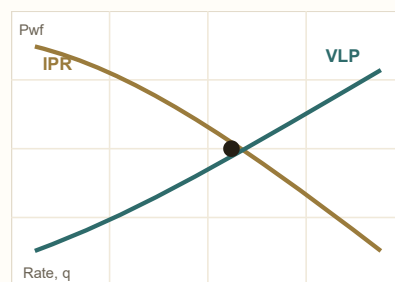


WellSim

Nodal analysis, connected reserves and production forecast, for oil, water and gas wells — the engineering method of the M. El-Ashry workbooks, rebuilt as explicit, tested software.

NODAL ANALYSIS



Four modules, one well model every module reads the same calibrated well

01

Well model — nodal analysis

The operating point for oil, water and gas wells under natural flow, gas lift or a full ESP design — pump matching, stage and wear back-calculation, and a design proof against the curve.

123 ESP pumps · 4 catalogues

Producer + injector

02

Reserve estimate

Connected volumes by material balance — Havlena–Odeh, static-gauge and reservoir-limit for oil; p/Z from production data, SITHP or memory gauges for gas, condensate on a total-gas basis.

STOIIP · GIIP

 p/Z tank

03

Production forecast

Saturation-tracking material balance coupled to the nodal model — Turner and Walsh's generalized MB with volatilized oil R_v ; history and forecast on one calendar axis.

Turner

Walsh (R_v)

EUR

04

Artificial-lift selection

Screen ESP, gas lift, sucker rod, jet pump and PCP against their envelopes across the well's life, then rank the survivors by UDC economics — the cheapest applicable method.

ESP

Gas lift

Rod

Jet

PCP

Explicit, traceable, and tested the physics you can point at

345 43/43 9+ digits

unit tests,
all green

golden-case
sweep

workbook parity against
the source workbooks

- Composite Vogel IPR with a Darcy-from-K default J, calibrated to a test.
- Brill & Beggs Z everywhere — explicit, no goal-seek.
- Well-head temperature always calculated by Ramey, not typed.
- Two pressure records — an anchor and a working reservoir pressure.
- Unit-aware boundaries — stb/d, MMscf/d, psi, kept canonical.
- Gas impurities are inputs wherever gas properties appear.

ON THE WEB

Open **wellssim.app** — an installable app that keeps working offline once loaded.

PORTABLE APP

A single signed **.exe** — no install, USB-friendly, each case a folder beside it.

YOUR DATA STAYS YOURS

Free use needs no account; Save and Open keep every case a file you control.